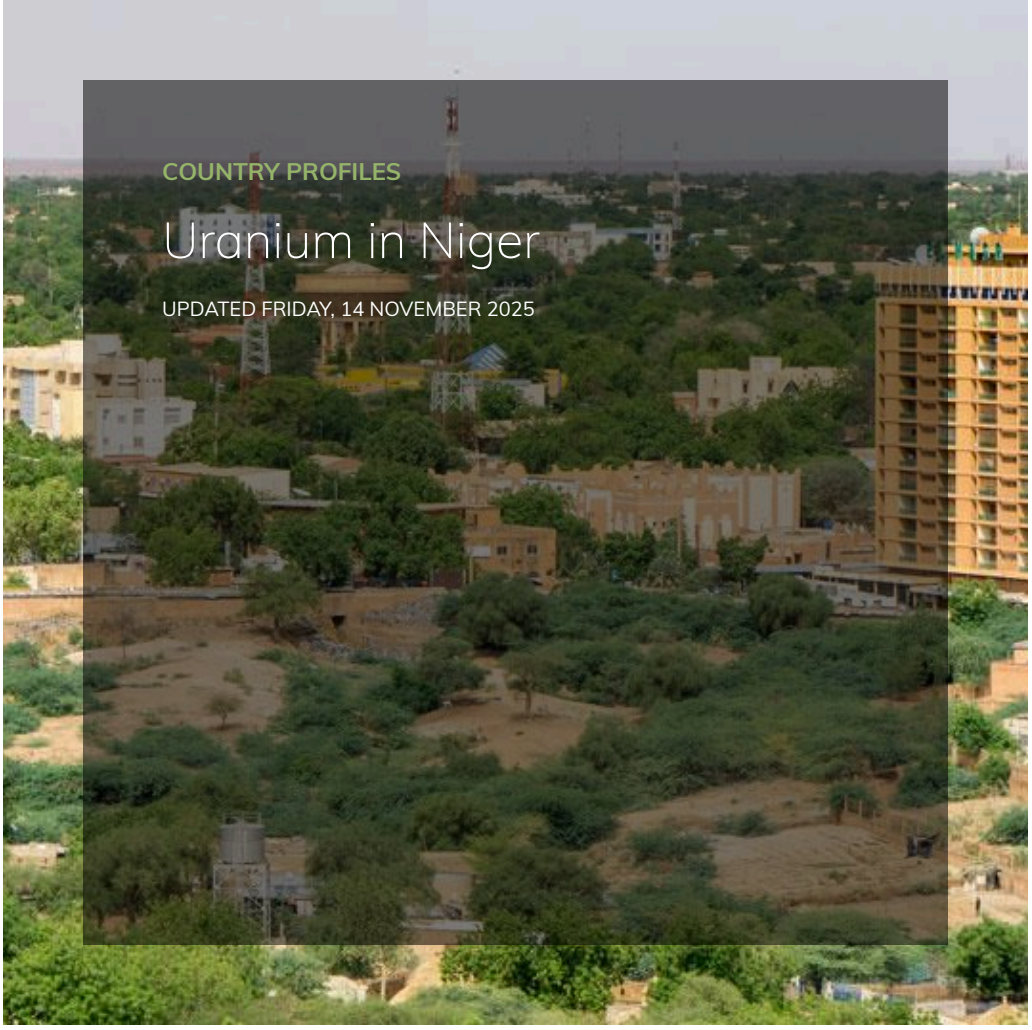

COUNTRY PROFILES

Uranium in Niger

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- Niger has two significant uranium mines providing about 5% of world mining output from Africa's highest-grade uranium ores.
- Niger's first commercial uranium mine began operating in 1971.
- In July 2023 Niger's government was overthrown. Since then, operating permits for a number of mines have been withdrawn.

Privacy - Terms

Uranium was discovered at Azelik in Niger in 1957 by the French Bureau de Recherches Geologiques et Minières (BRGM), looking for copper. The French Atomic Energy Commission (CEA) initiated further studies. Further discoveries in sandstone followed including at Abokurum (1959), Madaouela (1963), Arlette, Ariege, Artois & Tassa/Taza (1965), Imouraren (1966) and Akouta (1967). In the midst of this, Niger became independent of France in 1960.

In 1964 the coal deposit of Thirozerine was also discovered. It is currently operated by SONICHAR and produces electricity for the northern Agadez region, including the uranium mines.

Historically, uranium mining in Gabon has been closely linked with Niger due to the role of the French Atomic Energy Commission and Cogema (now Orano).

Uranium mining today

In 2023 Niger's total identified uranium resources were about 454,000 tU, including 315,000 reasonably assured resources and 139 inferred resources.

Niger has a cumulative uranium production of 158,889 tU from 1945-2024, ranking it 9th globally.

Uranium is mined close to the twin mining towns of Arlit and Akokan, 900 km northeast of the capital Niamey (more than 1200 km by road) on the southern border of the Sahara desert and on the western range of the Air mountains. The concentrates are trucked 1600 km to Parakou in Benin, then railed 400 km to Cotonou port and exported for conversion, mostly to Comurhex in France.

Production is first sold to the partners in proportion to their equity at an 'extraction price' determined by the government, notionally based on operation costs, but somewhat higher. The partners then sell or use it, in the case of the government, through a trading company.

In July 2023 Niger's government was overthrown. The leaders of the military coup have since revoked the mining permits of Orano at Imouraren and GoviEx at Madaouela.

SOMAIR: Arlit/Arlette, Tamou, Tagora, Artois

SOMAIR is 63.4% owned by Orano and 36.6% by Office National des Ressources Minières du Niger (ONAREM) through Sopamin, the

Niger mining assets company. Orano is the operator.

The Société des Mines de l'Aïr (SOMAIR) was formed in 1968 and started production from the Arlette/Arlit deposit in 1971, by open cut mining of 0.30-0.35% ore down to 70 metres depth. The mine is 250 km north of Agadez and 7 km northwest of Arlit town. Capacity was subsequently expanded to about 2100 tU/yr in 1981 (though half was then laid up). Since 2003, production ramped up again, with the Tamou deposit producing 1565 tU in 2006. The Artois deposit is deeper (90 metres) and at a lower grade (0.20-0.25%). Mill capacity was increased to 3000 tU/yr in 2012 with sodium uranate as the product. Average head grade in 2015 was 0.28%U. Resources at end of 2016 are tabulated below. A 16% drop in production was planned by 2017 (from 2015).

A new 1.4 Mt per year heap leach operation for low-grade ore (<0.1%U) – Somair Lixi – has contributed up to 1000 tU/yr to production from 2010. Mine operations are certified under ISO 14001 for environmental management.

In May 2013 a terrorist car bomb damaged the mine plant and killed one employee, also injuring 14. Production partially resumed four weeks later, in mid-June, and was fully restored in August. Four French nationals including an Areva employee, among a group of seven who were kidnapped from Arlit in 2010, were released in October 2013.

In March 2020 Orano reported that it intended to ensure the continuity of uranium production in northern Niger, so expected Somair production to continue for ten years, or until the Imouraren project resumes.

At the end of 2013 Areva reported 20,000 tU inferred resources for a 100% owned Arlit concession.

Production in 2021 was 1996 tU with an average grade of 1.81%. There were plans in 2022 to decarbonize the Somair site through the construction of a photovoltaic plant.

COMINAK: Akouta, Akola, Afasto/Ebba

COMINAK is 59% owned by Orano, 10% by Spain's Enusa, and 31% by ONAREM through Sopamin. Orano is the operator.

The Compagnie Miniere d'Akouta (COMINAK) was set up in 1974 and started production from the Akouta deposit in 1978, a few kilometres southwest of Akokan, and then from Akola and Afasto

orebodies. This was an underground operation at a depth of about 250 metres, with 250 km of tunnels. Mill capacity was 2000 t/yr of magnesium uranate (75% U) or 1800 tU/yr. Head grade in 2015 was 0.4%U. There was a 30% drop in production from 2015 to 2018 and with reserves almost exhausted the mine closed in March 2021 after producing 75,000 tU.

Cominak took steps to improve its competitiveness. Production switched to the new deposit of Ebba/Afasto, south of Akouta and Akola. Mine operations are certified under ISO 14001 for environmental management.

In March 2021, the Akouta mine ceased production after more than 40 years of operation and 75,000 tU extracted.

SOMINA: Azelik

The Societe des Mines d'Azelik SA (SOMINA) is a joint venture established in 2007 to mine Azelik/Teguidda, 160 km southwest of Arlit and 150 km northwest of Agadez, in the Agadez region. Its equity is 37.2% China's CNNC International, 33% Niger government, 24.8% ZXJOY Invest (Chinese) and 5% Trendfield Holdings Ltd. In 2009 Trendfield sold its 5% of the Teguidda/Azelik deposit to Korea Resources Corp (KORES).

Azelik is being developed with major Chinese (CNNC) equity and came into production at the end of 2010, with the aim to ramp up to 700 tU/yr. It is an open pit and underground operation using alkaline leach, and with resources of 15,600 tU at 0.2%. CNNC said in August 2014 that Azelik has experienced prolonged project delays, overruns in its construction budget, and low production "which led to heavy losses and causing default [on the] repayment of bank loans." In February 2015 CNNC International announced that the mine would be closed and put on care and maintenance due to "tight cash flow". It had earlier hoped to raise production to 2500 t/yr by 2015 and double that by 2020.

Niger mine production (tonnes U)

	2011	2012	2013	2014	2015	2016	2017	2018	2021
SOMAIR	2726	3065	2730	2331	2509	2164	2116	1783	1996
COMINAK	1075	1506	1508	1501	1607	1313	1332	1128	190
SOMINA	64 est	96 est	290 est	225 est	0	0	0	0	0
Total	3865	4667	4528	4057	4116	3477	3449	2911	2186

Imouraren

Imouraren Inc. is 66.65% owned by Orano Expansion (86.5% Orano, 13.5% Korea Electric Power Co (Kepco/KHNP)) and 33.35% by the state: Niger government (10%) and Sopamin (23.35%). Kepco paid €170 million for its share and the right to take 10% of production. This is in line with Kepco's 2.5% equity in the Georges Besse II enrichment plant in France. In February 2012, in connection with a 20,000 tU purchase agreement over 15 years, EdF agreed with Areva (now Orano) to take a 12.7% stake in the mine but this did not proceed. In January 2013 CNNC's SinoU subsidiary agreed to buy a 10% stake for €200 million, though this did not proceed.

Development of the large Imouraren deposit about 80 km south of Arlit and 160 km north of Agadez was confirmed in January 2008, after Areva (now Orano) agreed to increase royalty payments to the government by 50%, following a 2006 agreement. In January 2009 Orano was awarded a mining licence.

The Imouraren project is a €1.9 billion investment, and Orano has also agreed to spend €6 million per year on health, education, training, transport and access to water and energy for local people. Orano was aiming for initial production in 2014, ramping up to 5000 tU/yr for 35 years. The deposit covers 8 km by 2.5 km and Orano lists 213,700 tonnes of uranium reserves at 0.07%U, plus 62,500 tU indicated resources. Orebody depth is between 100 and 150 metres and maximum thickness is 60 m.

Excavation of the first pit was under way in mid-2012, but labour disputes put the schedule in sufficient doubt for the government to warn the company that delays were unacceptable. Areva (now Orano) agreed to pay the government €35 million to support security at the site. In May 2014, with uranium prices not sufficient to justify mining of the deposit, the government and Areva (now Orano) agreed to set up a joint strategic committee to determine when mining should start. Almost €1 billion in capital expenditure was still required, and 4 billion tonnes of ore and overburden will need to be moved over the life of the mine.

In 2015-16 Orano recorded €194 million and then €316 million impairment charges on the project, writing it down to €348 million.

An earlier Imouraren joint venture agreement was signed in 1974 but development stalled on economic grounds.

In September 2022 Orano confirmed that optimization studies assessing the “feasibility of operating the deposit with other extraction methods” were under way for the Imouraren project.

In June 2023 Nigerien authorities withdrew Orano's operating permit for the mine.

New mines and prospects

Madaouela

Goviex Uranium's Madaouela project is 15 km from the Arlit and Akouta mines (SOMAIR & COMINAK) in the Arlit region of the Air Massif, and was discovered by the CEA in the early 1960s. The Madaouela project comprises five deposits (Marianne, Marilyn, Miriam, Madaouela South North East [MSNE] and Maryvonne). Trendfield (25%) and UK-based GoviEx Uranium formed the GoviEx Niger JV in 2007 to explore the Madaouela and Anou Melle mineralization, but Trendfield then exchanged this equity for a 10% share of GoviEx. GoviEx Uranium is a private company founded by Govind Friedland, which holds 10% of it following an IPO in 2014. Denison holds 20% of GoviEx Uranium, after transferring some African interests in 2016. Toshiba, Ivanhoe Industries and Cameco are also shareholders, with 9%, 6% and 4% respectively. The Niger government holds a 10% free carried interest and in April 2019 acquired another 10% for $14.5 million. In April 2012, Toshiba Corporation entered a convertible debt-financing agreement with GoviEx Uranium, providing support for the company's operations through to the start of uranium extraction and processing, from which time it has a 14-year offtake agreement for 230 tU/yr.

In November 2017, measured and indicated resources of the Madaouela uranium project were 42,600 tU plus 10,640 tU inferred resources (NI 43-101 compliant), most in sandstone of the Marianne/Marilyn (M&M), Miriam and MSNE (Madaouela South North East) deposits over 15 km. The unusual Miriam deposit was discovered in 2012. Probable reserves were 23,330 tU, with underground mining of M&M and MSNE, open pit for Miriam. GoviEx completed an environmental impact and social impact assessment (ESIA) in March 2015, applied for a mining permit in June, and the government issued the mining permit in February 2016 for Madaouela I, an area including the above deposits. In April 2019 the government included Agaliouk in the Madaouela 1 mining permit. It is adjacent to Madaouela 1 and covers part of Miriam as

well as MSEE (Madaouela South Extreme East) and La Banane deposits. As of 2022 Miriam proven and probable reserves were 4718 tU (4696 tU of which are proven). M&M, MSNE, Maryvonne underground deposits have a total of 17,255 tU with 3353 tU proven and 13,902 tU probable.

Conventional processing following radiometric sorting and use of ablation technology is expected to produce 1040 tU/yr over 21 years, with potential for expanding the resource. Ablation applies a physical, grain-size beneficiation process to ore slurries. Start-up capital cost is expected to be $359 million, and operating cost $24.50/lb U_3O_8, or $31.50 including royalties. Project economics are based on a $70/lb uranium price.

GoviEx's application for the Eralrar exploration tenement was approved in February 2016. The application for the Agaliouk (Agal) tenement, a significant part of the project, was not approved until November 2017. In April 2019 the government agreed to grant renewed nine-year permits for the Madaouela 2, 3&4, and Anou Melle exploration tenements.

In September 2022 Goviex published its feasibility study for Madaouela. The study revealed that Madaouela is one of the largest uranium resources in the world, with 100 million pounds of U_3O_8 in measured and indicated resources. The life of mine uranium production is estimated at 50.8 million pounds U_3O_8 (averaging 2.67 million pounds U_3O_8 per annum over 19 years). The total initial capital cost is $343 million.

In January 2023 GoviEx Uranium announced that 20 institutions that expressed an interest in financing the project were shortlisted for the due diligence phase. The project finance process is expected to take several months.

In July 2024 the country's Minister of Mines informed GoviEx of the removal of its mining rights.

Dasa

Global Atomic Corporation, a public company listed on the Toronto Stock Exchange, is developing the Dasa project, part of the Adrar Emoles tenement in the Tim Mersoi basin – about halfway between Arlit and Agadez, and south of Orano's Cominak and Somair uranium mines.

In April 2020, Global Atomic announced the results of an updated preliminary economic assessment of its Phase 1 mining plan that

featured underground mining of the high-grade (>0.5%U) 'Flank Zone'. This study demonstrated that with a two-year build and a capital cost of $203 million (including 20% contingency) it would be expected to achieve a 26.6% internal rate of return (IRR) at a $35/lb assumed uranium price and an all-in sustaining cost of $18.39/lb U_3O_8. The Phase 1 plan provided 44 Mt of feed material at 5396 ppm to recover over 44 million lbs of U_3O_8 over 12 years of production from the Flank Zone. Subsequent mining will be from other mineralized horizons given the resource has been estimated at over 102 million pounds U_3O_8 indicated and 88 million pounds U_3O_8 inferred.

In December 2020 the government granted a mining permit for the project. Global Atomic plans to start building the mine in early 2022, with a view to commercial operation about the end of 2024.

In late 2021 a feasibility study reported “initial Phase 1 mine development over 12 years producing 45.4 million pounds”. Initial excavation was completed in mid-2022, along with significant infrastructure development.

Earlier in July 2017, the company signed an ore sales agreement with Orano Mining to sell a portion of its production to Orano for processing by Somair at Arlit, 100 km north. The company’s Tin Negouran (west of Agadez), Isakanan and Dajy (part of Adrar Emoles) deposits are less advanced, but do host non-NI 43-101 compliant uranium resources.

In February 2023 a local court in Agadez issued several orders against Global Atomic Corporation's mining subsidiary SOMIDA (Société Minière de Dasa S.A.), including an order to cease all operations under the Adrar Emoles 4 exploration permit. Global Atomic said the orders would not affect the Dasa project, which is covered by the Adrar Emoles 3 permit.

An updated resource estimate was published by Global Atomic in May 2023, showing 109.6 million pounds of U_3O_8 at 0.49%, up 50% from a July 2019 estimate of 73.1 million pounds U_3O_8 at 0.53%.

In October 2023 the government expressed its "full support" for the project.

As of May 2023 Dasa has indicated resources of 42,040 tU and inferred 19,770 tU.

Abokorum

In July 2006 the China National Nuclear Corporation (CNNC) agreed to develop the 12,790 tU **Abokorum** deposit in the Agadez region, through its subsidiary China Nuclear International Uranium Corporation (SinoU), but no more has been heard of this.

SinoU and China's ZTE Energy Corporation have established a joint venture to carry out uranium exploration near the Azelik mine.

Trendfield formed the UREX joint venture (approx 50:50) with Australia's Artemis Resources to explore the **Tagaza** deposits adjacent to Teguidda. (Parent company Trendfield Energy and Resources is a China-based "private international mining and consulting firm".)

In April 2007 the government issued uranium exploration permits to Areva (now Orano), Rio Tinto and others for the Tchirozérine area, 40 km northwest of Agadez. An Indian company took out an exploration licence in the Arlit region.

In January 2022 Niger's government granted four exploration permits to Tejia Resources for uranium in the Arlit and Tchirozérine departments of the Agadez region.

Niger Uranium Reserves and Resources

	Reserves	Resources		
	proven & probable	measured	indicated	inferred
SOMAIR	36,700 tU @ 0.12%		21,115 tU @ 0.11%	30,311 tU @ 0.14%
Arlit concesion				12,845 tU @ 0.16%
COMINAK	88 tU @ 0.38%	116 tU @ 0.32%	100 tU @ 0.32%	
Imouraren	171,336 tU @ 0.08%		22,368 tU @ 0.07%	6475 tU @ 0.07%
SOMINA Azelik		9,684 tU @ 0.14%		
Madaouela		11,568 tU @ 0.1%	25,705 tU @ 0.1%	7535 tU @ 0.11%

	Reserves	Resources		
Dasa			39,080 tU at 0.15%	33,695 tU @ 0.15%

NEA and IEA Uranium 2022 Resources, Production and Demand

As reported in the 2024 *Red Book*, Niger's total identified uranium resources (reasonably assured and inferred) amount to approximately 336,000 tU up to $130/kgU, with almost all (98%) accessible by open pit.

In April 2007 the government said that it aimed for uranium production of 10,500 tU/yr "in the next few years", and named Areva as its strategic partner in uranium development.

Areva was reported to have been paying royalty on the basis of a product valuation of 27,300 CFA francs ($57) per kilogram, and in 2007 this was increased to 40,000 CFA ($83/kg), plus the provision of 300 tonnes of product for Niger to sell on the open market. This was then sold to Exelon in the USA for $42 million.

In August 2008 Niger Uranium Ltd announced an inferred resource of 1700 tU at **In Gall**, this being Samrec-compliant and in shallow sandstone.

In January 2010 NGM Resources announced an inferred resource of 5000 t U3O8 at Takardeit, some 100 km south of Imouraren. Paladin Energy made a $24 million takeover bid for NGM, but it decided to let this lapse in October 2010 due to armed hostilities in the region. However, the Australian Takeovers panel disallowed the decision and Paladin proceeded with the takeover of NGM and its Indo Energy Ltd (IEL) subsidiary, in line with the recommendations of NGM directors. Following attacks on Areva's operations, in mid-2013 Paladin ceased its exploration activities and invoked a *force majeure* consideration.

In 2009 Korea Resources Corp. agreed to buy 400 tonnes per year of uranium or U_3O_8 and take a 5% share of the Teguidda mine in central Niger from Trendfield, a Chinese company.

Non-proliferation

Niger is party to the Nuclear Non-Proliferation Treaty. It has a comprehensive safeguards agreement in force and in 2004 signed the Additional Protocol.

Notes & references

General sources

Orano
OECD-NEA & IAEA, *Uranium 2022: Resources, Production and Demand* (often referred to as the 'Red Book')
WISE